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# 1.) Importing the necessary libraries.
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# 2.) Loading and cleaning the dataset.
df = pd.read_csv('/content/train (1).csv')
df['Age'] = df['Age'].fillna(df['Age'].median())
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# 3.) Converting the text into numbers.
df['Sex'] = df['Sex'].map({'male': 1, 'female': 0})

# 4.) Select the features (inputs) and the target to be predicted.
features = ['Pclass', 'Sex', 'Age', 'SibSp', 'Parch']
X = df[features]
y = df['Survived']
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# 5.) Splitting the dataset.
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 6.) Initializing and training the model.
model = RandomForestClassifier()
model.fit(X_train, y_train)
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▾ RandomForestClassifier
RandomForestClassifier()
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# 7.) Checking the accuracy.
predictions = model.predict(X_test)
print(f"Model Accuracy: {accuracy_score(y_test, predictions):.2%}")
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Model Accuracy: 81.01%
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